

MARK YANG

(609)-580-0118 | yang.mar@northeastern.edu | Boston, MA | www.linkedin.com/in/markyang05 | <https://mar-yan24.github.io/>

EDUCATION

Northeastern University

Boston, MA (April 2028)

- BS in Electrical and Computer Engineering (3.98 GPA)
- Relevant Coursework: Embedded Design, Circuits and Signals, Fundamentals of Networks, Neuromechanical Simulation of Human Movement (MS level class), Differential Equations, Physics 2
- Activities: Putnam Club, Rock Climbing Club, Tau Beta Pi Engineering Honors Society

Queens University Belfast

Belfast, Northern Ireland (September–December 2024)

- N.U.*in* Study Abroad Semester

PROFESSIONAL EXPERIENCE

Philips Cambridge NA HQ

Software Test Engineer Co-op

Cambridge, MA (Jan 2026–Present)

- Defined feature behavior for the PIC iX event notification system by writing Gherkin tests with the Solution Reliability team, producing 10+ test cases and Gherkin's used for validation and verification.
- Built an internal RAG LLM within the Azure cloud pipeline to automate feature vulnerability collation, cutting manual defect analysis time by 60–70% and delivering real-time test coverage recommendations.

NEUMove

Undergraduate Researcher

Boston, MA (Aug 2025–Present)

- Programmed the compliant muscle tendon unit (CMTU) in MuJoCo alongside a PhD collaborator and currently fitting the CMTU to existing OpenSim models for integration with the lab's direct collocation repository.
- Developing GPU-accelerated reinforcement learning pipelines using autodifferentiation in MuJoCo.Warp to rapidly train humanoid musculoskeletal models for locomotion tasks.

Northeastern University

Engineering Center Aide

Boston, MA (Aug 2025–Jan 2026)

- Supported 900+ freshman engineering students as a hardware and software assistant, teaching and debugging Raspberry Pi Pico, Python, MATLAB, AutoCAD, and Solidworks projects.
- Maintained makerspace 3D printers and laser cutters, advised students on print settings and material selection, and hosted bi-weekly soldering workshops.

Synbio Technologies

Software Engineering Intern

South Brunswick, NJ (Jun–Aug 2024)

- Developed a backend JavaScript event-logging system to track user interactions and auto-generate usage plots, enabling the team to pinpoint and resolve UX bottlenecks.
- Overhauled 100+ pages of the company website using behavior data from the logging system to prioritize redesign targets, boosting user engagement with interactive features post-launch.

PROJECTS

Short Horizon Actor Critic Musculoskeletal Rapid Imitation Learning

Research Project (Mar 2026–Present)

- Collaborating with the DeepMind MuJoCo team and Nvidia Warp team to implement determinism and auto-differentiation (AD) support in MJWarp.
- Testing SHAC RL policy training on MJWarp with cartpole and ant actors to later implement a rigid 22-muscle myoelg model for GPU-accelerated locomotion training.

Bipedal Spring Loaded Inverse Pendulum (bSLIP) Runner

Graduate Class Project (Sep 2025–Jan 2026)

- Implemented a two-leg bSLIP runner in MATLAB/Simulink (Simscape Multibody) with stance/swing PD control and contact/friction modeling.
- Performed apex-to-apex Poincare return-map analysis and added a one-step deadbeat regulator via touchdown hip-angle control to maintain apex height on rough terrain.

RoboCup Junior Maze Simulated Robot

Group Project (Dec 2021–Jul 2022)

- Implemented a Breadth-First Search (BFS) navigation algorithm on the Erebus simulation platform and built custom maze worlds to systematically debug edge cases.
- Placed 1st in the 2022 RCJ international competition hosted in Thailand.

SKILLS AND HONORS

Languages/Tools Python, Java, C, C++, MATLAB, Simulink/Simscape, Git, Verilog, LTSpice, AWS

Awards/Honors 1st in Simulation Division 2022 RCJ Thailand, Morgan Stanley Early Insights